United International University

Department of Computer Science & Engineering

Mid-term Exam, Trimester: Summer 2026

Course Code: CSE 3715 **Course Title:** Data Communication

Duration: 1 hour 30 minutes **Total Marks:** 30

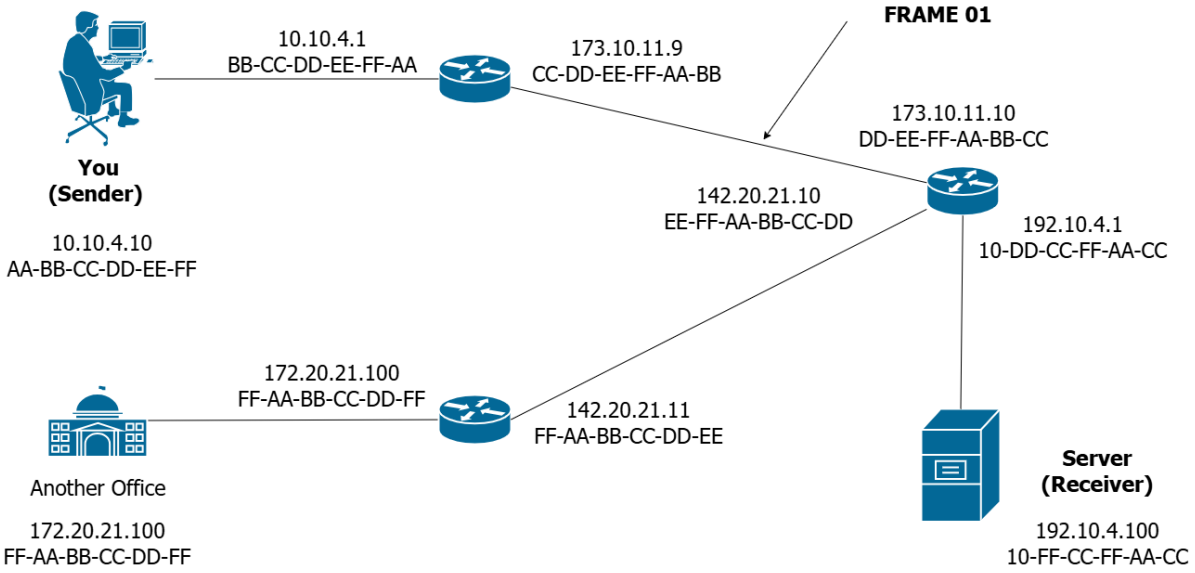
(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

ANSWER ALL THE QUESTIONS

1. Suppose, you have joined a government office as a network administrator after completing your graduation. Your office is setting up a new network. Inside the office, they want a network that is **secure** and **fault-tolerant** for their employees (cost is not a problem). For connecting to the internet, they are deciding between using **fiber optic cable** with a national ISP or using **Starlink**.

   (a) Which network topology would you suggest for the internal network, and why? **[2]**

   (b) The office chooses fiber optic for external communication. Data is sent to a server abroad through the internet gateway. The diagram below shows the path with several routers in between, along with the frame format. What will be the values of Frame 01, based on the given frame format? Assume the necessary port numbers. **[3]**



Destination MAC	Source MAC	Source IP	Destination IP	Source Port	Destination Port	Data

Table 1: Frame Format (Simplified)

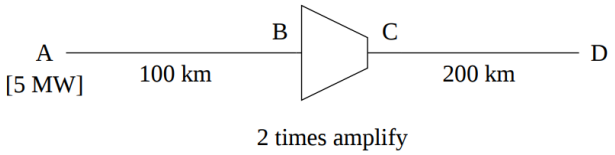
   (c) Assume, you used twisted pair cable inside the office for internal communication. Why are the wires twisted, and how does this improve the signal? **[2]**

   (d) Some officials in your office do not understand Starlink. Using a diagram, explain how Starlink provides internet access through wireless communication. **[3]**

2. A signal is generated with a power of 100 mW and first passes through a **noisy channel** with a noise power of 10 μW and bandwidth of 15 kHz.

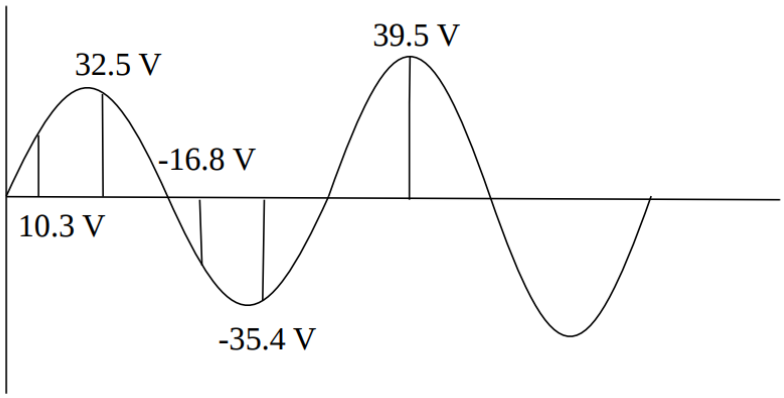
   (a) **Calculate** the channel capacity of the noisy channel, and **determine** the minimum number of signal levels required for transmission if the system utilizes only 70% of the channel capacity. **[5]**

   (b) Since the destination is far away, the signal is boosted to 5 MW and sent over a long-distance **wire** shown below, entering at point A. The power loss rate from A to B is 5 kW/km, and from C to D is 0.05 dB/km. **Calculate** the total change in signal power in decibels, and comment on whether the signal is amplified or attenuated overall. **[5]**



3. Suppose you have some analog data (e.g., voice) that needs to be sent over a network. Regardless of whether the network is **Wi-Fi** (wireless) or **Ethernet** (wired), the analog data must first be encoded into digital form using **Pulse Code Modulation (PCM)**. After this, Wi-Fi modulates the encoded data and sends it over the wireless medium, while Ethernet line codes the same encoded data and sends it directly over the wired medium.

   (a) The figure below shows a sampled analog signal. Using PCM with 8 quantization levels, find the quantized value and encoded code at each time stamp. Assume amplitudes range from -40V to +40V. **[4]**



   (b) To send this data over **Wi-Fi**, it must first be modulated. Using the first 6 bits of your encoded data (e.g., 101 111), draw the resulting waveform using: **[2]**
   (i) Binary Amplitude Shift Keying
   (ii) Binary Phase Shift Keying

   (c) To send this data over **Ethernet** instead, it must be line coded. Using the first 6 bits of your encoded data (e.g., 101 111), draw the resulting signal using: **[4]**
   (i) NRZ-I
   (ii) Manchester Encoding